

HUSSAM AZHAR

Software Engineer • Licensing, DRM & Payment Systems • Full-Stack & Hybrid Desktop

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Portfolio: hussamazhar.github.io/hussam-portfolio • LinkedIn: linkedin.com/in/hussam-azhar-64845b122

PROFESSIONAL PROFILE

Software Engineer with 5+ years delivering scalable web applications, cross-platform hybrid desktop software and RESTful APIs for a US commercial software publisher — with deep, hard-to-source specialization in Digital Rights Management, software licensing and subscription payment systems. Sole owner of a production DRM platform that today manages 4.3M+ license keys across 352,000+ paying customers, and architect of the license-key, entitlement and renewal flows embedded in flagship CAD and creative products shipping on Windows, Mac, Linux and Chromebook. Fluent across React.js, Node.js, TypeScript, C#/.NET, PHP and C++, with a track record of taking features from database schema to production deployment independently and of leading small teams to do the same. Known for clean, maintainable code, measurable performance gains and a low defect return rate from QA.

CORE SPECIALISATION — LICENSING, DRM & PAYMENTS

- **DRM platform ownership** — designed, built and continue to own the iDRM Portal end to end: license-key generation, activation, validation, revocation and entitlement enforcement for IMSI Design LLC's entire commercial product line — 4.3M+ keys, 352,000+ customers in live production.
- **Client-side enforcement** — developed the C++ DRM client SDK embedded in TurboCAD, handling secure activation handshakes, offline grace periods and tamper-resistant validation against the portal.
- **Floating & multi-seat licensing** — implemented floating-license workflows allowing customers to share and reclaim seats across multiple machines without over-provisioning revenue.
- **Subscription billing** — integrated Braintree and PayPal for in-app subscription renewal inside TurboCAD, and engineered the bridge between WooCommerce storefronts and the DRM portal so purchases, renewals and refunds reconcile automatically.
- **Order fulfilment services** — built and deployed Windows and web services (C#, .NET, Entity Framework, SQL) that fulfil online-store orders with keys issued by the DRM portal, eliminating manual key delivery.
- **Lifecycle automation** — JWT-secured REST APIs and webhooks drive cron-scheduled email-drip campaigns (PHP, PHPMailer) that notify customers of expiry, renewal and refund events, protecting recurring revenue.
- **Commercial visibility** — delivered a CRM portal and analytics dashboard giving sales and support instant lookup of orders, keys and customer records, plus live sold-product and refund statistics.

TECHNICAL SKILLS

Licensing & Payments	DRM architecture, license-key lifecycle, floating licenses, entitlement enforcement, Braintree, PayPal, WooCommerce, JWT, Webhooks
Front-End	React.js, Next.js, React Hooks, Context API, Redux, TypeScript, JavaScript ES6, Material UI / MUI v5, amCharts, Chart.js, ApexCharts, HTML5, CSS3, SASS, Bootstrap, jQuery, Vue.js
Back-End	Node.js, Express.js, Python, C# / ASP.NET (MVC & Core), Entity Framework, PHP, Laravel MVC, C++, RESTful API design, Axios, Firebase Auth
Desktop	Electron.js (Main + Renderer, API & CLI), Electron.NET.API, WPF, Swift (printing module), Mac App Store submission & code signing
Databases	MS SQL Server, MySQL, MongoDB, SQLite, Firestore, BigQuery, ERD design, schema modelling, query optimization
Cloud & DevOps	Google Cloud (Cloud Functions, Cloud Run, Cloud Tasks, BigQuery, Pub/Sub, Cloud Storage, Firestore, Cloud Scheduler, Cloud Build), MS Azure & Azure App Service, Docker, Nginx, GitHub Actions CI/CD, Windows/Web service deployment, WinSCP, XAMPP
Tools	Git, SmartGit, Visual Studio, VS Code, Azure DevOps/TFS, Xcode, Unity (WebGL builds), JIRA, Postman, Fiddler, Figma
Practices	Clean code, performance tuning, requirement analysis, R&D, technical leadership, stakeholder reporting

PROFESSIONAL EXPERIENCE

Software Engineer

November 2021 – Present

IMSI Design LLC — California, USA (Remote)

- Develop and maintain three flagship CAD products — hybrid desktop applications shipping on Windows, Mac, Linux and Chromebook OS, built in Electron.js, React.js, Redux, TypeScript, Material UI and Node.js.
- Own the company's Digital Rights Management and product-licensing portal (PHP, MySQL, JWT), continuously extending it with new features while it manages 4.3M+ license keys and 352,000+ customer records in production.
- Engineered the C++ DRM client SDK and floating-license workflow inside TurboCAD, giving customers seat-sharing across machines while protecting license integrity.
- Integrated Braintree into WooCommerce storefronts and built the bridge to the DRM portal for automated subscription-license renewal; implemented in-app PayPal renewal directly inside TurboCAD.
- Built and deployed Windows and web services (C#, .NET, Entity Framework, SQL) that serve online-store orders to customers with DRM-generated license keys, removing manual fulfilment overhead.
- Automated customer licensing communications with PHP, PHPMailer and cron-scheduled drip campaigns covering expiry, renewal and refund events.
- Delivered an internal CRM portal (PHP, MySQL, JWT) for instant order, license-key and customer lookup by sales and support teams.
- Manage production databases and handle deployment of Windows and web services across company servers.
- Lead R&D on new feature sets to extend product capability, and provide the strongest technical resolution to functional anomalies raised by QA.
- Handle Mac App Store submissions, including packaging, signing and compliance review.
- Present weekly progress across all active projects directly to the CEO and stakeholders.

Associate — Digital & App Innovation, Engineering

July 2020 – November 2021

Prismware Technologies — Islamabad, Pakistan (Remote)

- Built serverless data pipelines on Google Cloud, ingesting third-party API data into BigQuery via Cloud Functions (Node.js) to enable fast, interactive analysis of stored datasets.
- Developed and deployed event-driven serverless applications integrating Cloud Functions with Pub/Sub, Cloud Storage and Firestore to automate data workflows without server management.
- Scheduled recurring jobs with Cloud Scheduler to trigger automated backups, reporting and system maintenance.
- Implemented secure APIs connecting React and Node.js applications to backend services, validated end to end with Postman.
- Converted Figma designs into production-ready React components (Hooks, Redux) and contributed new feature sets to existing products.
- Diagnosed problems across web projects and advised the team on the most robust solution path.

Associate Software Engineer

January 2018 – June 2020

IT-Falcons (Pvt.) Ltd. — Lahore, Pakistan

- Delivered three web projects from scratch through to deployment using .NET MVC architecture.
- Integrated the SOG e-commerce payment gateway into a Laravel MVC project — first exposure to the payment-systems work that became a career specialization.
- Resolved defects across WPF C# desktop applications, MVC C# web applications and WordPress theme templates (PHP) powering company storefronts.
- Gathered requirements directly from clients, performed analysis and recommended optimal development approaches.
- Supported senior engineers on database and back-end development; tracked and reported defects through JIRA.

SELECTED PROJECTS

Full portfolio with case studies, screenshots and architecture notes: hussamazhar.github.io/hussam-portfolio

iDRM Portal | IMSI Design LLC

Digital Rights Management and licensing platform built from scratch and still in production, managing approximately 4,321,431 license keys across 352,273 customers. Handles key generation and validation, a JWT-secured REST API, webhook-driven email notifications and an admin dashboard surfacing live sales and refund statistics. Flow: customer purchase → key generated and issued via REST API → client app validates against the portal (JWT) → renewal and refund events fire webhooks → automated cron-scheduled email drips notify the customer.

Stack: PHP 8, MySQL, JWT, REST APIs, Webhooks, PHPMailer, Bootstrap, jQuery, HTML5/CSS3

TurboCAD — Windows | IMSI Design LLC | www.imsidesign.com/products/turbocad/turbocad-windows

Professional 2D/3D CAD suite for experienced users, offering hundreds of drawing tools, an optional AutoCAD-like drafting interface, 3D modelling, photorealistic rendering and extensive file support. Developed the C++ DRM client SDK that communicates with the iDRM Portal, implemented the floating-license workflow for seat-sharing across a customer's machines, and integrated Braintree and PayPal for in-app subscription renewal.

Stack: C++, DRM Client SDK, Braintree, PayPal, Floating Licensing

The Print Shop for Mac | IMSI Design LLC | www.encore.com/the-print-shop-for-macintosh

Desktop design studio with 10,000+ royalty-free professional backgrounds and 400+ templates, plus photo-editing tools to import, crop and colour-correct personal images. Owned the front-end, SQLite schema, Node.js back-end, license validation against the iDRM Portal and the Mac App Store submission.

Stack: React (JSX, Hooks, Context, Redux), TypeScript, Material-UI, SASS, Electron (Main + Renderer), Node.js, SQLite, Axios

PrintMaster v9 — Windows & Mac | IMSI Design LLC | www.encore.com/printmaster-v9

Cross-platform creative suite for greeting cards, invitations, postcards, calendars and business brochures, with thousands of royalty-free templates and no license limitations on output. Built the React front-end, the .NET Core + Electron.NET back-end, the SQLite data layer, license activation against the iDRM Portal and the Swift printing module.

Stack: React, TypeScript, Redux, .NET Core 2.2, ElectronNET.API, Node.js, SQLite, Swift

Mavis Beacon Teaches Typing — Web | IMSI Design LLC | mymavisbeacon.com

Browser-based edition of one of the world's most recognized typing tutors, rebuilt as a MERN platform that serves two audiences from a single codebase — structured professional mastery for teens, adults and the workplace, and game-driven learning for younger users. Delivered 100+ skill checks and timed typing tests, personal speed and accuracy goal setting, real-world typing challenges attached to every lesson, posture and ergonomics video instruction, and an automatically issued completion certificate at eight hours of practice. Built the React front-end and the Node.js/Express REST API with MongoDB persistence, and engineered the game layer — embedding Unity WebGL builds alongside 12+ knowledge-based games and a wizard challenge written natively in React with animation-driven interactions.

Stack: React, Redux, Advanced UI Frameworks, React Animations, Node.js, Express.js, MongoDB (MERN), Unity WebGL, REST APIs

Brainy RMS — Web (SaaS) | Prismware Technologies | www.brainyrms.com

Smart revenue-management SaaS platform that converts PMS reservations, competitor and market rates, seasonality signals and local-event data into optimal daily room pricing for hotels. Built the React + TypeScript dashboard end to end — daily and monthly pricing views, BI and pace reports, sales statistics, smart-budget and revenue-target configuration, a multi-property admin panel and interactive API documentation — alongside the serverless Python and Node.js back-end running on Google Cloud behind Firebase authentication.

Stack: React 18, TypeScript, Redux, Material-UI, amCharts / Chart.js / ApexCharts, Firebase Auth & Firestore, Python, Node.js, Google Cloud Functions, BigQuery, Pub/Sub, Cloud Tasks, Cloud Run, Docker

SLOPE PMS Integration (Brainy RMS) — Google Cloud | Prismware Technologies | www.brainyrms.com

Serverless integration service that continuously synchronizes lodging reservations from the SLOPE property-management system into Brainy's analytics warehouse, keeping automated pricing decisions aligned with live booking data. Engineered the cloud function end to end — token-authenticated paginated REST polling, incremental last-update filtering, duplicate detection against BigQuery, batched inserts and Pub/Sub events that trigger downstream price recalculation.

Stack: Node.js, JavaScript, Axios, Moment.js, Google Cloud Functions (Functions Framework), BigQuery, Cloud Pub/Sub, dotenv

SLYDE — Web (companion site for iOS & Android apps) | Prismware Technologies | www.slydeline.com

Public-facing launch and compliance website for the SLYDE mobile applications. Built the Next.js + React front-end on a custom Express server, delivering an MUI-themed responsive landing experience together with full Terms of Service and Privacy Policy pages, containerized with Docker and Nginx and shipped through GitHub Actions CI/CD to Azure App Service, with Google Cloud Build configured as an alternate deployment path.

Stack: Next.js 12, React 17, MUI v5 / Material-UI v4, Express, Node.js, Firebase, Docker, Nginx, GitHub Actions, Azure App Service, Google Cloud Build

Star Marketing | IT-Falcons (Pvt.) Ltd. | www.starmarketingonline.com

Website and property sale-purchase panel for one of Pakistan's leading real-estate marketing firms. Customers browse cost-effective residential and commercial listings across Karachi, Lahore, Islamabad and Peshawar, post their own property ads and manage them from a personal dashboard. Delivered the admin panel, customer dashboard, sale/purchase portal and Pusher-powered live chat.

Stack: Laravel 5.5 (PHP MVC), Vue.js, jQuery, MySQL, Bootstrap, Pusher

BAZ Consulting | IT-Falcons (Pvt.) Ltd. | bazconsulting.com

Corporate site for an advisory firm connecting entrepreneurs, corporate organizations and NGOs to accelerate delivery of sustainable solutions. Delivered the responsive front-end with a clean corporate presentation.

Stack: HTML5, CSS3, Bootstrap, JavaScript, jQuery

EDUCATION

Government College University — Lahore, Pakistan <i>BS (Hons.) Computer Science Major: Web Engineering & Databases</i>	2013 – 2017
Punjab College of Science — Lahore, Pakistan <i>FSc. Pre-Engineering</i>	2011 – 2013
Qurban & Surraiya Educational Institute — Lahore, Pakistan <i>Matriculation in Science</i>	2009 – 2011

CERTIFICATIONS

.NET Specialist <i>EVenture Solutions (EVS) Training Institute</i>	Reg. No: L0116060052
Microsoft Azure <i>MicrotechX-IT Training Institute</i>	Reg. No: 16-04-GCUL-409

ADDITIONAL INFORMATION

Strengths	Disciplined and deadline-driven; able to run a project independently or lead a team; strong client communication and requirement gathering; fast learner who adapts quickly to unfamiliar technologies.
Languages	English (professional working proficiency), Urdu (native), Punjabi (native)
Availability	Open to full-time roles and contract engagements. One month's notice in current position, potentially negotiable. References available on request.